

# VeriFake Response to Reviewers

IEEE CSDE 2026 | Paper ID ieee-csde\_3824

VeriFake: A TF-IDF and Ensemble Learning Framework for Web-Based Fake News Detection

We thank the reviewers for identifying the experimental and presentation issues. The revision replaces the earlier results with the completed corrected experiment, states the evaluation boundaries, and revises the conclusions accordingly. LinearSVC is selected by training cross-validation, and the paper reports the loss of performance under dataset transfer. The manuscript follows the supplied IEEE conference template with individual author blocks. Locations below refer to the six-page revised manuscript. Reviewer comments are reproduced as excerpts and divided into points where needed.

## Response to Reviewer 1

| Reviewer Comment                                                                                                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Please rerun the complete experiment using a leakage-safe pipeline in which the data are split before TF-IDF fitting.                                                                                              | <p>Action: The completed revision experiment splits articles before feature fitting. TF-IDF and each classifier are fitted inside the training CV folds. Vocabulary, document frequencies, and hyperparameters are never fitted using held-out test articles. We additionally removed exact body duplicates before splitting. The final random split contains 16,205 training and 4,052 test records. The completed repair run supplies the results reported in the revision.</p> <p><b>Location:</b> Section III-A and III-B, p. 2; Fig. 1, p. 3; Algorithm 1, p. 4.</p>                                         |
| 2. Verify the dataset's original class-label definitions and report any remapping or balancing explicitly.                                                                                                            | <p>Action: We retain 0 for reliable/real and 1 for unreliable/fake in Kaggle. The raw counts are 10,387 and 10,413, respectively. We removed 160 empty bodies, 159 repeated title-and-body records, and 224 additional repeated bodies. The cleaned counts are 10,387 real and 9,870 fake. No balancing, resampling, or Kaggle label remapping is used. For ISOT, True.csv is assigned 0 and Fake.csv is assigned 1. These labels remain dataset annotations.</p> <p><b>Location:</b> Section III-A, p. 2.</p>                                                                                                    |
| 3. Recalculate all metrics directly from the final predictions and ensure that Table II agrees exactly with the confusion matrices; ROC-AUC should use probability or decision scores rather than binary predictions. | <p>Action: The revised primary comparison is Table III. It uses saved predictions and continuous scores from the completed run. LinearSVC has TN=2,038, FP=40, FN=35, and TP=1,939, giving 98.15% accuracy and MCC 0.9630 on 4,052 articles. Fig. 2 reports the same counts. ROC-AUC is 0.9977 and is calculated from decision scores. Probabilistic classifiers use positive-class probabilities. We also report conditional bootstrap intervals and Holm-corrected exact McNemar tests.</p> <p><b>Location:</b> Section III-D, pp. 3-4; Table III, p. 4; Section IV-A, pp. 4-5; Fig. 2, p. 5; Fig. 3, p. 6.</p> |
| 4. Please also correct the RF, Decision Tree, LinearSVC, MLP, and TF-IDF mathematical descriptions;                                                                                                                   | <p>Action: We replaced the incorrect expressions with sublinear TF and smoothed IDF followed by L2 normalization; a linear squared-hinge objective with the implemented intercept treatment; Gini-based classification-tree splitting; averaged tree class probabilities for RF; and binary cross-entropy with L2 weight regularization for the MLP. The MLP hidden activation, sigmoid output, Adam settings, and early-stopping limits are stated. Its internal early-stopping subset limitation is also disclosed.</p> <p><b>Location:</b> Section III-C, pp. 2-3; Eqs. (1)-(4), p. 3.</p>                     |

## Response to Reviewer 1 continued

| Reviewer Comment                                                                                                                                                            | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5. provide the exact input text construction, split ratio, random seeds, hyperparameters, and optimization procedure;                                                       | <p>Action: The primary input is normalized title, one space, and body. The revision specifies normalization, tokenization, stop words, n-grams, document-frequency limits, feature cap, seed 42, outer splits, three-fold CV, and MCC selection. Table II lists every candidate grid. Fixed model settings, training limits, optimizer choices, and software versions are included. Forward transfer retains Kaggle training-CV settings; reverse transfer tunes on ISOT only. Input ablations freeze the title-and-body settings within each protocol.</p> <p><b>Location:</b> Section III-A to III-E, pp. 2-4; Tables I and II; Algorithm 1.</p>                                   |
| 6. replace the incorrect dataset reference;                                                                                                                                 | <p>Action: The unrelated polyp-segmentation reference has been removed. The revision identifies the Kaggle Fake News competition, the ISOT laboratory documentation, and Ahmed et al. for ISOT. The acquisition mirrors are identified separately from the original dataset sources. The bibliography has also been restricted to relevant dataset, method, and evaluation references.</p> <p><b>Location:</b> Section III-A, p. 2; Section III-E, p. 4; references [4], [14], and [15], p. 6.</p>                                                                                                                                                                                   |
| 7. and add source-disjoint/cross-dataset evaluation.                                                                                                                        | <p>Action: We report a known-author-disjoint evaluation and both transfer directions. LinearSVC accuracy is 96.80% on known-author-disjoint Kaggle, 71.49% from Kaggle to ISOT, and 58.47% in reverse. Tuning is confined to each training source, and exact cross-collection matches are removed. We explicitly state that normalized author-string separation is not publisher separation. Publisher-disjoint validation remains untested; the new results answer the cross-dataset part of the request without overstating the author proxy.</p> <p><b>Location:</b> Section III-B, p. 2; Section IV-B, p. 5; Table IV, p. 5; Fig. 2, p. 5; Fig. 3, p. 6; Section V, pp. 5-6.</p> |
| 8. The terms "fact checking" and "verification" should be moderated because the current model performs text classification rather than evidence-based factual verification. | <p>Action: We now describe VeriFake as supervised classification of dataset labels. The abstract and discussion state that it does not verify claims using external evidence. The interface description distinguishes the prototype from the exported pipeline and from evaluated behavior. We removed claims of a validated secure or real-time factual-verification service. LinearSVC margins are explicitly distinguished from probabilities of truth.</p> <p><b>Location:</b> Abstract and Section I, p. 1; Section III-E, p. 4; Section V, pp. 5-6; Fig. 3, p. 6.</p>                                                                                                          |
| 9. Finally, the novelty claim should be clarified, since the current work compares existing classifiers rather than introducing a new ensemble architecture.                | <p>Action: We agree. The contribution is the controlled comparison, evaluation of author-input sensitivity and dataset transfer, and documented preprocessing-aware export. The title is retained, but ensemble learning is explicitly defined as the established forests and boosting methods in the comparison. Algorithm 1 documents the implemented evaluation procedure and is not presented as a new learning algorithm. We make no state-of-the-art or first-of-its-kind claim.</p> <p><b>Location:</b> Sections I and II, pp. 1-2; Section III-D, pp. 3-4; Section VI, p. 6.</p>                                                                                             |

## Response to Reviewer 2

| Reviewer Comment                                                                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. The key novelty and principal contributions should be highlighted more clearly in the introduction and conclusion.                                                        | <p>Action: The introduction now states three concrete contributions: a common training-only comparison of thirteen classifiers, author and cross-dataset evaluations with input ablations, and a documented evaluation procedure with a preprocessing-aware export. The conclusion returns to these contributions and the transfer limitation instead of claiming a new architecture.</p> <p><b>Location:</b> Section I, p. 1; Section VI, p. 6.</p>                                                                                                                                                                            |
| 2. Recent studies relevant to the problem should be added to strengthen the literature review and demonstrate how the proposed work advances the current state of knowledge. | <p>Action: The bibliography now contains 22 references, including nine additions to the previous revision. Related Work covers TELLER, LLM semantics mining, human/generated-news mixtures, dual-granularity transfer, TripleFact, dataset bias, and misinfo-general. Foundational sources distinguish article labels, social context, and evidence-based verification; methods sources support leakage control and MCC. Each addition is cited for a specific purpose. We do not claim numerical superiority over unmatched studies.</p> <p><b>Location:</b> Section II, pp. 1-2; references [1], [2], and [8]-[13], p. 6.</p> |
| 3. The manuscript organization should be improved to avoid abrupt transitions and enhance overall readability.                                                               | <p>Action: We reorganized the manuscript into Introduction, Related Work, Data and Methods, Experimental Results, Discussion and Limitations, and Conclusion. Methods precede the corresponding results. The results move from the primary comparison to author separation, transfer, and input sensitivity. Generic algorithm descriptions and repeated accuracy summaries were removed.</p> <p><b>Location:</b> Sections I-VI, pp. 1-6.</p>                                                                                                                                                                                   |
| 4. The methodological description should provide more detail on the workflow, assumptions, settings, and evaluation procedure.                                               | <p>Action: Fig. 1 shows the training and evaluation sequence, and Algorithm 1 specifies selection, refitting, transfer, ablation, and export. Tables I and II give sample sizes and hyperparameter grids. We state the label convention, normalization, group construction, CV selection rule, continuous-score AUC, and statistical procedures. We also distinguish author separation from publisher separation and explain the MLP early-stopping qualification.</p> <p><b>Location:</b> Section III, pp. 2-4; Tables I and II; Fig. 1; Algorithm 1.</p>                                                                      |

## Response to Reviewer 2 continued

| Reviewer Comment                                                                                                             | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5. The results should be organized more clearly and benchmarked against suitable existing approaches wherever possible.      | <p>Action: Table III provides all thirteen existing classifiers on the same held-out articles and representation, ordered by training CV MCC. LR, Naive Bayes, SVC, KNN, trees, ensembles, and MLP serve as common-protocol baselines. Table IV isolates generalization for the selected LinearSVC family, and Table V isolates input sensitivity. We discuss changed transfer rankings and avoid comparing unrelated published percentages as if they were evaluated on our test set.</p> <p><b>Location:</b> Section IV, pp. 4-5; Tables III-V.</p>                                                                                                                        |
| 6. The discussion should be extended to interpret the major findings, limitations, and significance of the obtained results. | <p>Action: The discussion explains the gap between random-split and transfer performance, the author-plus-title pattern, and possible topic, source-style, and annotation differences. It identifies the single outer split, limited search grids, exact rather than exhaustive near-duplicate control, missing-author exclusion, and untested publishers, temporal shifts, calibration, and live deployment. We also state that nonsignificant paired tests do not establish equivalence.</p> <p><b>Location:</b> Section IV-A to IV-C, pp. 4-5; Section V, pp. 5-6.</p>                                                                                                    |
| 7. Some figures should be regenerated at higher resolution and with clearer text and labels.                                 | <p>Action: The workflow was redrawn as four connected grayscale panels: prepare data, select models, test generalization, and evaluate results. It explicitly confines TF-IDF fitting and selection to training data. The revised workflow and result figures use vector graphics. The confusion matrix panels show article counts with labeled true and predicted axes. The ROC figure uses saved continuous scores and distinct line styles. We inspected all six compiled pages for readable text, labels, equations, and tables. The earlier word cloud and outdated result graphics were removed.</p> <p><b>Location:</b> Fig. 1, p. 3; Fig. 2, p. 5; Fig. 3, p. 6.</p> |
| 8. The captions of figures and tables should be sufficiently descriptive to communicate their content independently.         | <p>Action: Captions now state the dataset or protocol, model-selection rule where relevant, test size, positive class, metric units, and the meaning of matrix colors and ROC lines. Table V defines the random and author-disjoint abbreviations and the frozen-setting ablation. The figures and tables can therefore be interpreted without relying on the earlier manuscript captions.</p> <p><b>Location:</b> Captions for Tables I-V and Figs. 1-3, pp. 2-6.</p>                                                                                                                                                                                                       |

## Response to Reviewer 3

| Reviewer Comment                                                                                                                                 | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. The paper would benefit from a more explicit summary of its novelty and key contributions.                                                    | <p>Action: We added a specific contribution paragraph covering the shared comparison, transfer and author-input analysis, and documented export. We also explain that the evaluated classifiers already exist and that the evaluation algorithm describes the implemented procedure. The conclusion emphasizes what the experiments establish and the limits that remain.</p> <p><b>Location:</b> Section I, p. 1; Section III-D, pp. 3-4; Section VI, p. 6.</p>                                                                                                             |
| 2. A small number of recent and closely related references could be added to strengthen the literature review and better frame the research gap. | <p>Action: Seven studies from 2024-2026 now frame explainability, entity semantics, generated-news sensitivity, cross-domain transfer, contamination, dataset bias, and out-of-distribution evaluation. Nine references were added to the previous revision, for 22 in total, with numbered citations supporting the relevant claims. The discussion connects these studies to the author-input ablation and bidirectional transfer; irrelevant earlier references remain removed.</p> <p><b>Location:</b> Section II, pp. 1-2; references [1], [2], and [8]-[13], p. 6.</p> |
| 3. Minor improvements to the organization and section-to-section transitions would enhance the overall flow.                                     | <p>Action: The revision now follows a single sequence from the research question through related work, data preparation, fitting, evaluation, results, limitations, and conclusions. Subsections separate the four protocols from model settings and separate the primary comparison from transfer and input sensitivity.</p> <p><b>Location:</b> Sections I-VI, pp. 1-6.</p>                                                                                                                                                                                                |
| 4. Further clarification of selected methodological steps would improve transparency and reproducibility.                                        | <p>Action: The revision specifies input construction, original labels, cleaning counts, group-string handling, seeds, fold structure, TF-IDF settings, candidate grids, and optimizer settings. Algorithm 1 records when settings are selected, retained, or refitted. Section III-E reports the software environment, acquisition mirrors, repository URL, and the separation between evaluated models and the all-data deployment export.</p> <p><b>Location:</b> Section III, pp. 2-4; Tables I and II; Algorithm 1.</p>                                                  |

## Response to Reviewer 3 continued

| Reviewer Comment                                                                                                                      | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5. The results section could be made easier to follow by adding clearer comparisons with relevant baseline or recent methods.         | <p>Action: The thirteen existing methods now appear in one primary table under identical data and representation conditions. The selected model is determined by training CV MCC, even though LR has one fewer test error. We report paired tests, the change in model rankings under transfer, and separate generalization and input-ablation tables. Recent neural methods are discussed as context because no matched runs of those methods are available.</p> <p><b>Location:</b> Section II, pp. 1-2; Section IV, pp. 4-5; Tables III-V.</p> |
| 6. The discussion could provide somewhat more interpretation of the observed results and their implications.                          | <p>Action: We explain why 98.15% random-test accuracy and 71.49% forward-transfer accuracy support different claims. The author-plus-title result is interpreted cautiously because the test populations differ. The text also notes that adding authors to title plus body improves the group result slightly, so metadata effects are not uniformly harmful. Practical conclusions are limited to text classification and a documented export.</p> <p><b>Location:</b> Section IV-B and IV-C, p. 5; Section V, pp. 5-6.</p>                     |
| 7. Some figures may be improved through higher resolution and clearer labels.                                                         | <p>Action: We replaced the outdated result graphics with vector confusion matrices and a score-based ROC figure from the completed prediction records, and redrew the vector workflow as four connected grayscale panels with descriptive process names and a training-only selection boundary. Text and axes were inspected in the compiled six-page PDF. No additional model training was needed to prepare these presentation changes.</p> <p><b>Location:</b> Fig. 1, p. 3; Fig. 2, p. 5; Fig. 3, p. 6.</p>                                   |
| 8. The figure and table captions could be revised so that their main message is understandable independently of the surrounding text. | <p>Action: Each caption now defines the evaluation setting and the quantities shown. The confusion caption identifies true and predicted axes, counts, row-normalized colors, and model selection. The ROC caption identifies the positive class, continuous decision scores, AUC values, and chance line. Table captions specify percentage and unitless measures.</p> <p><b>Location:</b> Captions for Tables I-V and Figs. 1-3, pp. 2-6.</p>                                                                                                   |